

Seneca

SEG800

Digital Signal and Image Processing

The z-Transform

Seneca College

Overview

- The (direct) z-Transform
- Properties of the z-Transform
- Rational z-Transforms
 - Poles and Zeros
- The inverse z-Transform
 - Power series
 - Partial Fraction Expansion
- Analysis of LTI systems in the z-domain

Why z-Transform?

- Discrete-time equivalent of Laplace Transform of continuous-time signals
- It is important since
 - Allows for implementation of convolution as multiplication.
 - Characterizes an LTI system graphically by its pole-zero locations.

The Laplace Transform

- Two-sided (bilateral) Laplace Transform of a continuous-time signal:

$$X(s) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-st} \cdot dt, \quad \text{where } s = \sigma + j\omega$$

- One-sided (unilateral) Laplace Transform:

$$X(s) = \int_0^{+\infty} x(t) \cdot e^{-st} \cdot dt, \quad \text{where } s = \sigma + j\omega$$

The z-Transform

- Definition:

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$X(z) \equiv Z\{x(n)\}$$

$$x(n) \xleftrightarrow{z} X(z)$$

- The region of convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.

Figure 3.1.1 Region of convergence for $X(z)$ and its corresponding causal and anticausal components.

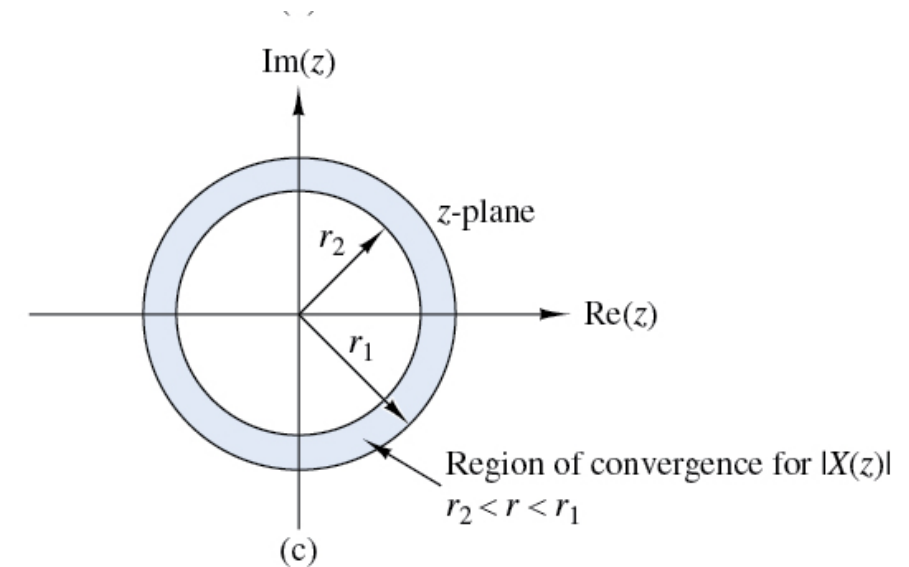
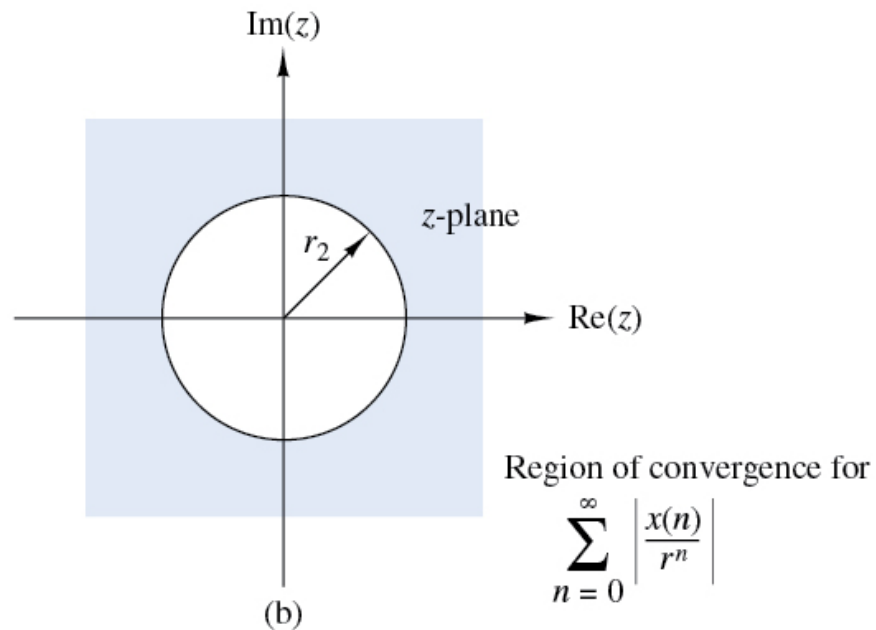
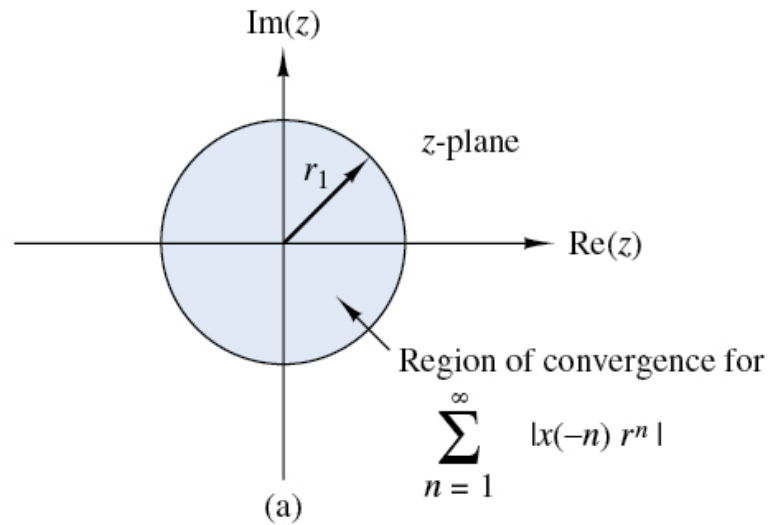


Figure 3.1.2 The exponential signal $x(n] = \alpha^n u(n)$ (a), and the ROC of its z-transform (b).

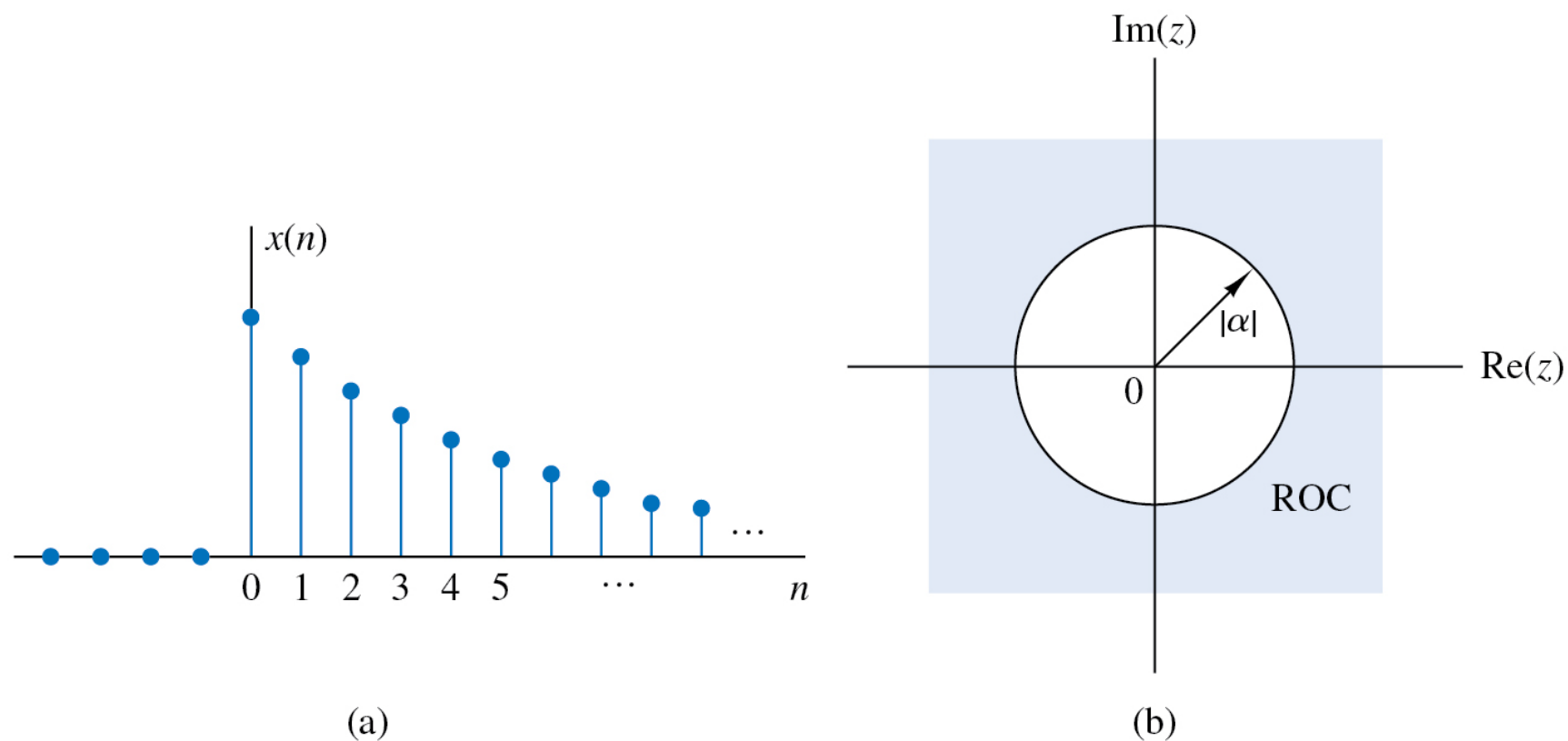


Figure 3.1.3 Anticasual signal $x(n) = -\alpha^n u(-n - 1)$ (a), and the ROC of its z-transform (b).

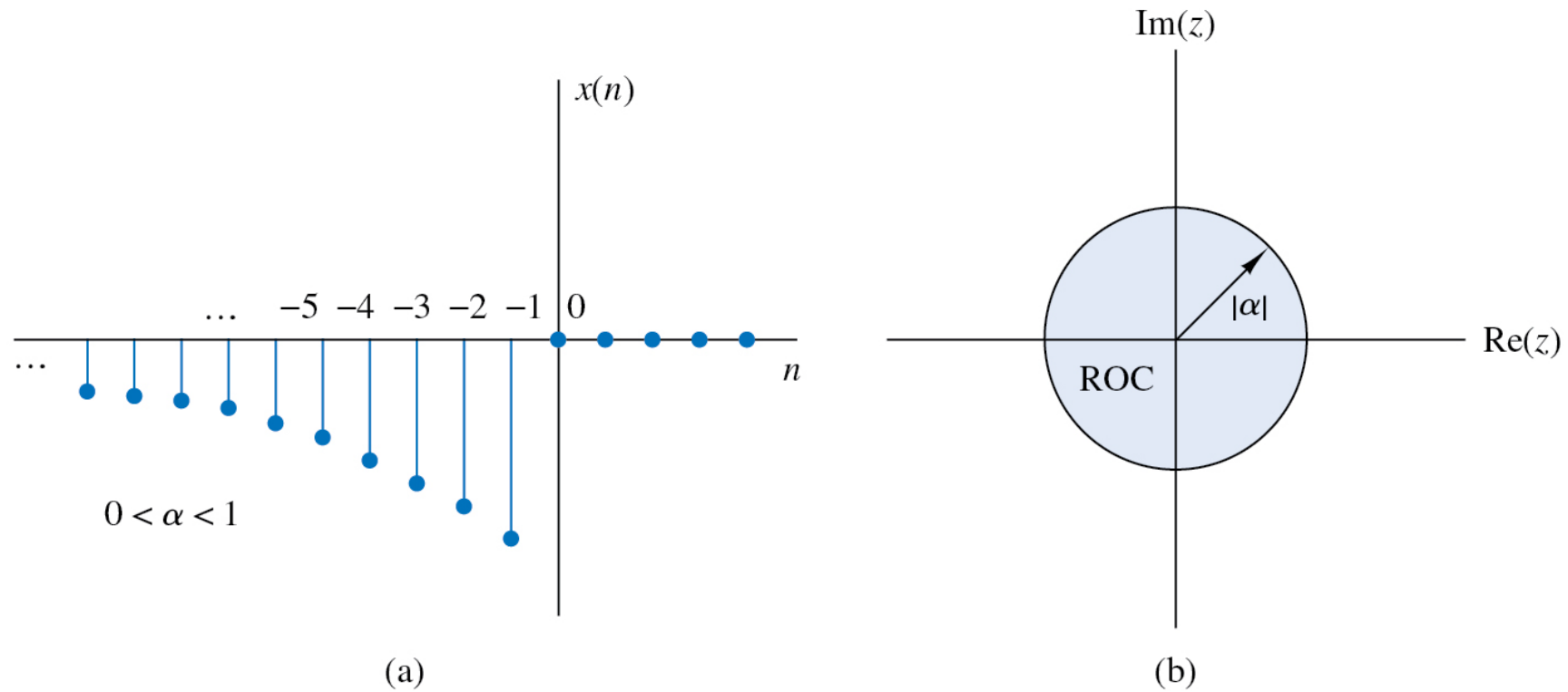


Figure 3.1.4 ROC for z-transform in Example 3.1.5.

$$x(n) = \alpha^n u(n) + b^n u(-n - 1)$$

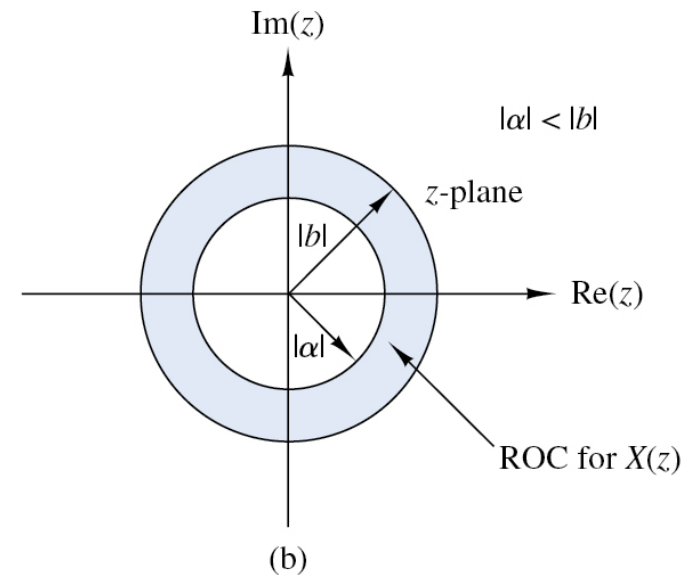
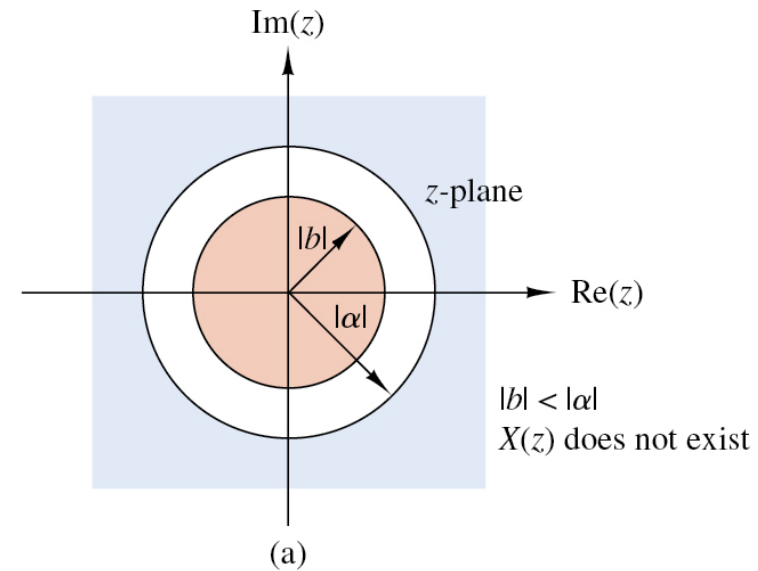
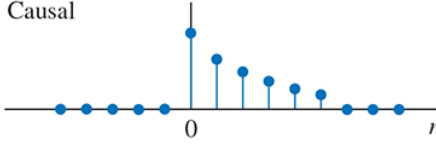
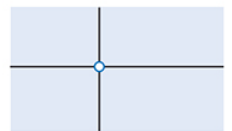
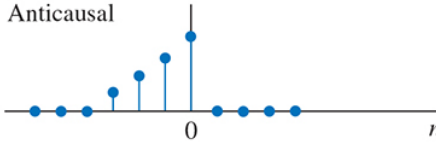
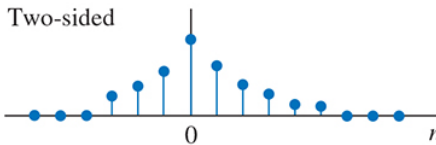
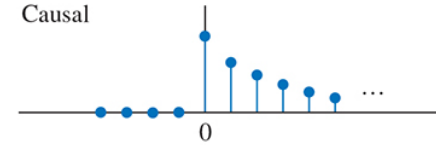
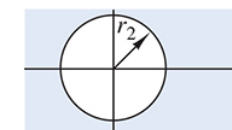
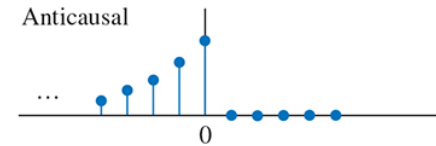
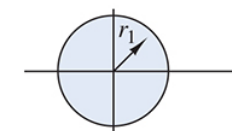
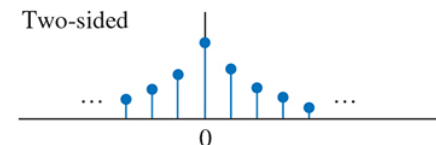


Table 3.1 Characteristic Families of Signals with Their Corresponding ROCs

Signal	ROC
Finite-Duration Signals	
Causal 	 Entire z-plane except $z = 0$
Anticausal 	 Entire z-plane except $z = \infty$
Two-sided 	 Entire z-plane except $z = 0$ and $z = \infty$
Infinite-Duration Signals	
Causal 	 $z > r_2$
Anticausal 	 $z < r_1$
Two-sided 	 $r_2 < z < r_1$

Properties of the z-Transform

Table 3.2 Properties of the z-Transform (page 1)

Property	Time Domain	z-Domain	ROC
Notation	$x(n)$	$X(z)$	ROC: $r_2 < z < r_1$
	$x_1(n)$	$X_1(z)$	ROC ₁
	$x_2(n)$	$X_2(z)$	ROC ₂
Linearity	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(z) + a_2X_2(z)$	At least the intersection of ROC ₁ and ROC ₂
Time shifting	$x(n - k)$	$z^{-k}X(z)$	That of $X(z)$, except $z = 0$ if $k > 0$ and $z = \infty$ if $k < 0$
Scaling in the z-domain	$a^n x(n)$	$X(a^{-1}z)$	$ a r_2 < z < a r_1$
Time reversal	$x(-n)$	$X(z^{-1})$	$\frac{1}{r_1} < z < \frac{1}{r_2}$
Conjugation	$x^*(n)$	$X^*(z^*)$	ROC
Real part	$\text{Re}\{x(n)\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	Includes ROC
Imaginary part	$\text{Im}\{x(n)\}$	$\frac{1}{2}j[X(z) - X^*(z^*)]$	Includes ROC

Table 3.2 Properties of the z-Transform (Page 2)

Property	Time Domain	z-Domain	ROC
Notation	$x(n)$	$X(z)$	ROC: $r_2 < z < r_1$
	$x_1(n)$	$X_1(z)$	ROC ₁
	$x_2(n)$	$X_2(z)$	ROC ₂
Differentiation in the z-domain	$nx(n)$	$-z \frac{dX(z)}{dz}$	$r_2 < z < r_1$
Convolution	$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$	At least, the intersection of ROC ₁ and ROC ₂
Correlation	$r_{x_1x_2}(l) = x_1(l) * x_2(-l)$	$R_{x_1x_2}(z) = X_1(z)X_2(z^{-1})$	At least, the intersection of ROC of $X_1(z)$ and $X_2(z^{-1})$
Initial value theorem	If $x(n)$ causal	$x(0) = \lim_{z \rightarrow \infty} X(z)$	
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi j} \oint_C X_1(v)X_2\left(\frac{z}{v}\right)v^{-1}dv$	At least, $r_{1l}r_{2l} < z < r_{1u}r_{2u}$
Parseval's relation	$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi j} \oint_C X_1(v)X_2^*(1/v^*)v^{-1}dv$		

Table 3.3 Some Common z-Transform Pairs (Page 1)

	Signal, $x(n)$	z -Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3	$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
4	$na^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
5	$-a^n u(-n - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $

Table 3.3 Some Common z-Transform Pairs (Page 2)

	Signal, $x(n)$	z-Transform, $X(z)$	ROC
6	$-na^n u(-n-1)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
7	$(\cos \omega_0 n)u(n)$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
8	$(\sin \omega_0 n)u(n)$	$\frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
9	$(a^n \cos \omega_0 n)u(n)$	$\frac{1 - az^{-1} \cos \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
10	$(a^n \sin \omega_0 n)u(n)$	$\frac{az^{-1} \sin \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

Convolution Property

- To determine the output of an LTI system
 1. Find the system function $H(z) = Z\{h(n)\}$ (done once)
 2. Find $X(z) = Z\{x(n)\}$ for input
 3. Compute $Y(z) = H(z) X(z)$
 4. Determine the inverse z-transform of $Y(z)$ to obtain $y(z)$

Rational z-Transforms and Pole-zero plots

Rational z-Transforms

- An important family of z-transforms:

$$\begin{aligned} X(z) &= \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \dots + a_N z^{-N}} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \\ &= \frac{b_0 z^{-M}}{a_0 z^{-N}} \frac{z^M + (b_1/b_0)z^{M-1} + \dots + b_M/b_0}{z^N + (a_1/a_0)z^{N-1} + \dots + a_N/a_0} \\ &= \frac{b_0}{a_0} z^{-M+N} \frac{(z - z_1)(z - z_2) \dots (z - z_M)}{(z - p_1)(z - p_2) \dots (z - p_N)} = G z^{N-M} \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)} \end{aligned}$$

Poles and Zeros

- The zeros of a z-transform $X(z)$ are the values of z for which $X(z) = 0$.
- The poles of a z-transform are the values of z for which $X(z) = \infty$.
- The ROC of a z-transform does not contain any poles.

Poles and Zeros for rational z-transforms

- A rational z-transform has
 - M finite zeros (z_1 to z_M) – roots of numerator polynomial
 - N finite poles (p_1 to p_N) – roots of denominator polynomial
 - $|N-M|$ zeros (if $N > M$) or $|M-N|$ poles (if $N < M$) at $z = 0$
 - And possibly zeros and poles at ∞
- Represented graphically
 - zeros by o and poles by x

$$X(z) = Gz^{N-M} \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)}$$

Figure 3.3.1 Pole–zero plot for the causal exponential signal $x(n) = a^n u(n)$.

$$x(n) = a^n u(n), \quad a > 0$$

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad \text{ROC: } |z| > a$$

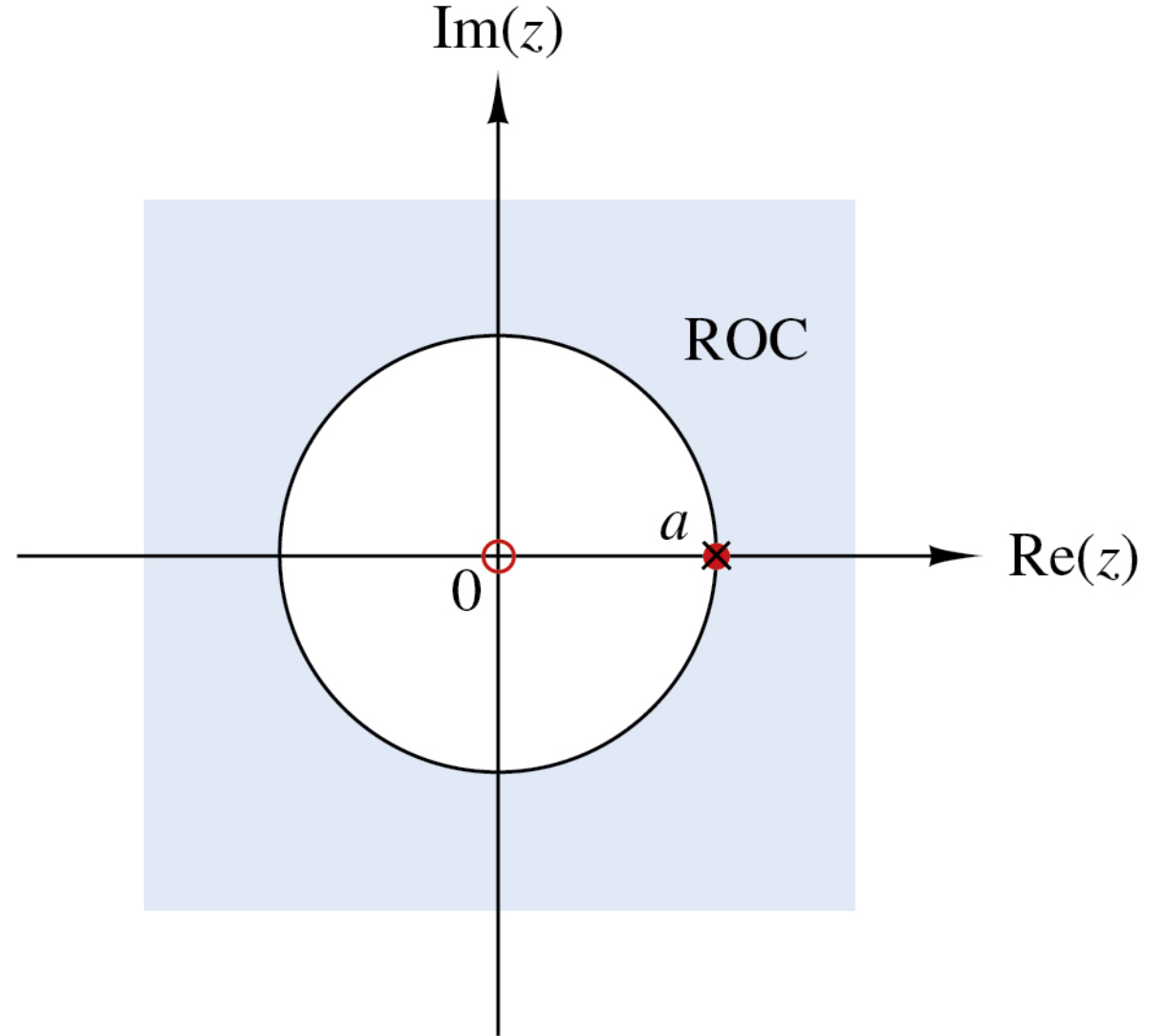


Figure 3.3.2 Pole–zero pattern for the finite-duration signal $x(n) = a^n$, $0 \leq n \leq M - 1$ ($a > 0$), for $M = 8$.

$$x(n) = \begin{cases} a^n, & 0 \leq n \leq M - 1 \\ 0, & \text{elsewhere} \end{cases}$$

where $a > 0$.

$$X(z) = \sum_{n=0}^{M-1} (az^{-1})^n = \frac{1 - (az^{-1})^M}{1 - az^{-1}} = \frac{z^M - a^M}{z^{M-1}(z - a)}$$

Since $a > 0$, the equation $z^M = a^M$ has M roots at

$$z_k = ae^{j2\pi k/M} \quad k = 0, 1, \dots, M - 1$$

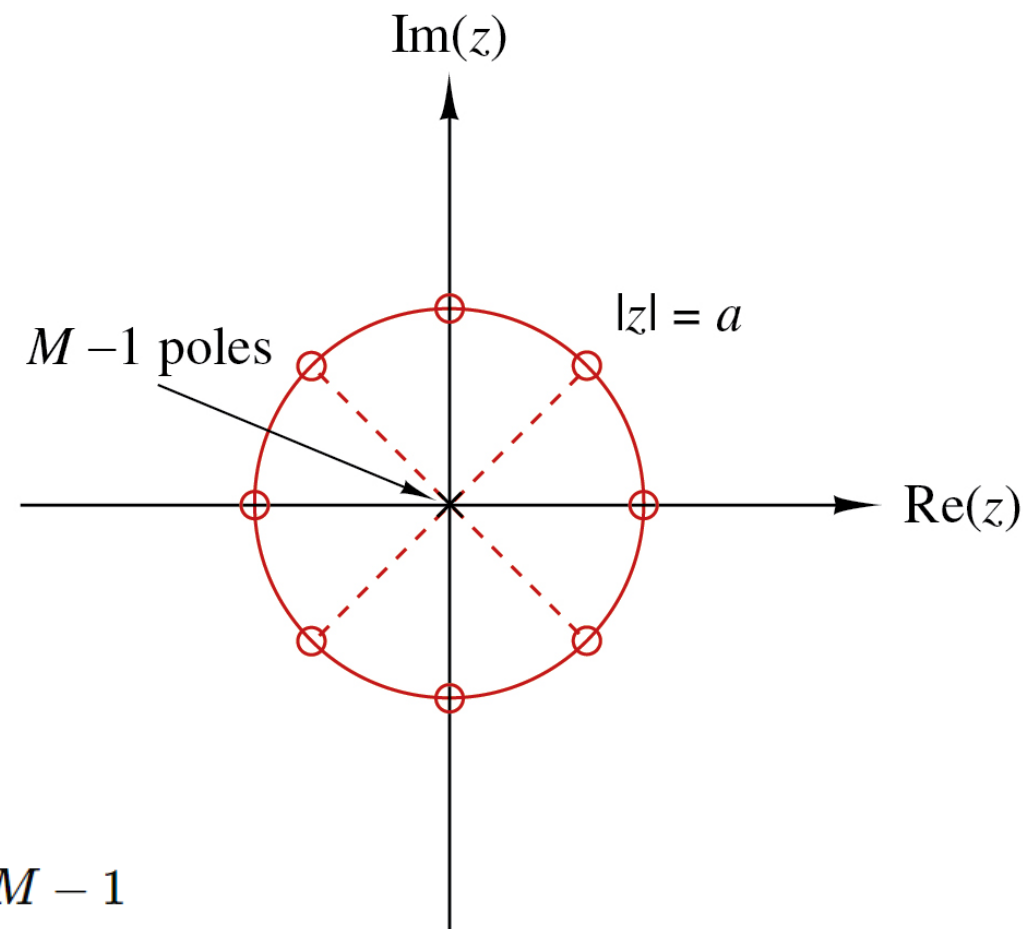


Figure 3.3.3 Pole-zero pattern for Example 3.3.3.

$$z_1 = 0, z_2 = r \cos \omega_0$$

$$p_1 = re^{j\omega_0}, p_2 = re^{-j\omega_0}$$

$$\begin{aligned} X(z) &= G \frac{(z - z_1)(z - z_2)}{(z - p_1)(z - p_2)} \\ &= G \frac{z(z - r \cos \omega_0)}{(z - re^{j\omega_0})(z - re^{-j\omega_0})}, \\ &= G \frac{1 - rz^{-1} \cos \omega_0}{1 - 2rz^{-1} \cos \omega_0 + r^2 z^{-2}}, \end{aligned}$$

$$\text{ROC: } |z| > r$$

$$\text{ROC: } |z| > r$$

Therefore,

$$x(n) = G(r^n \cos \omega_0 n)u(n)$$

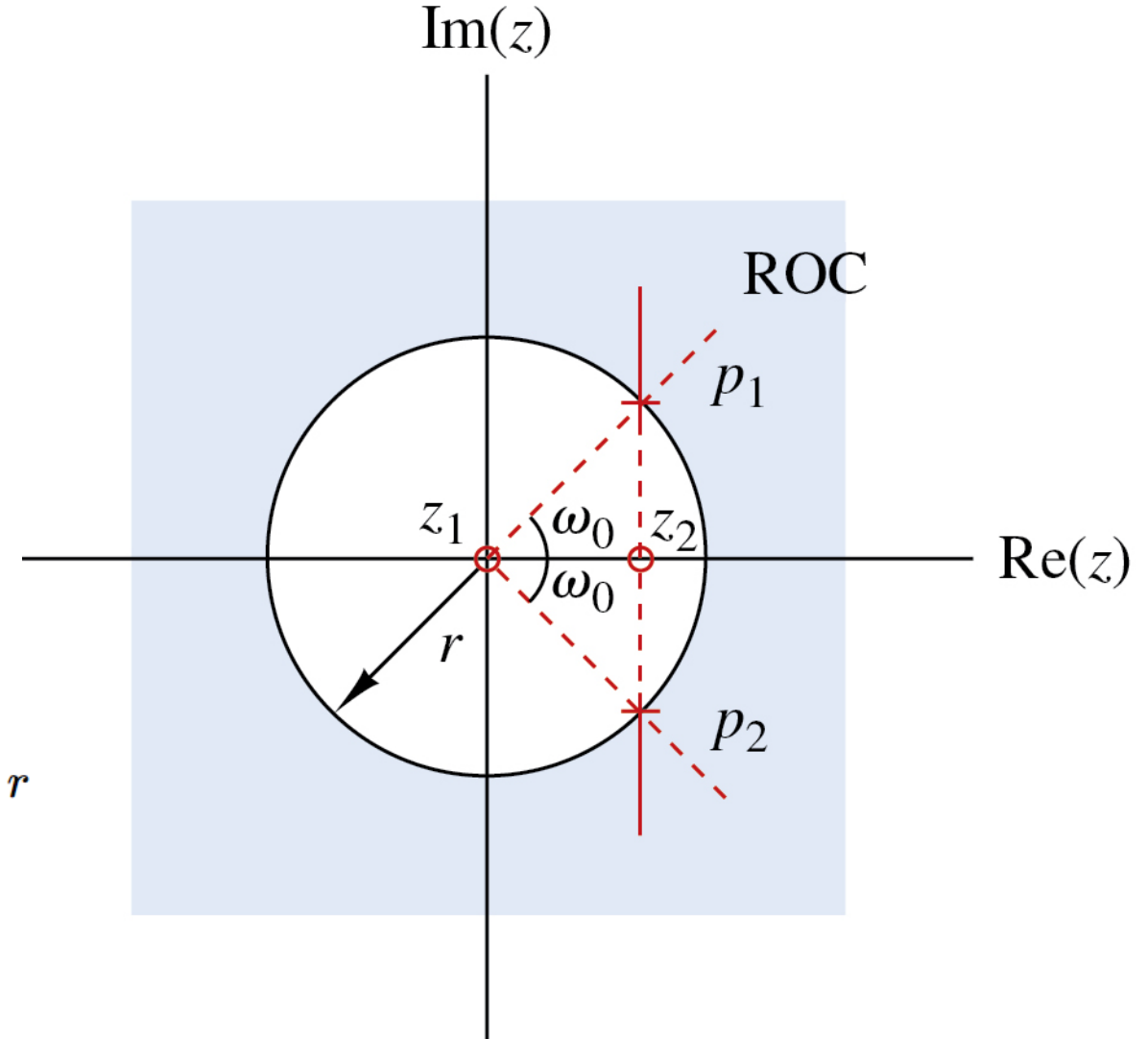
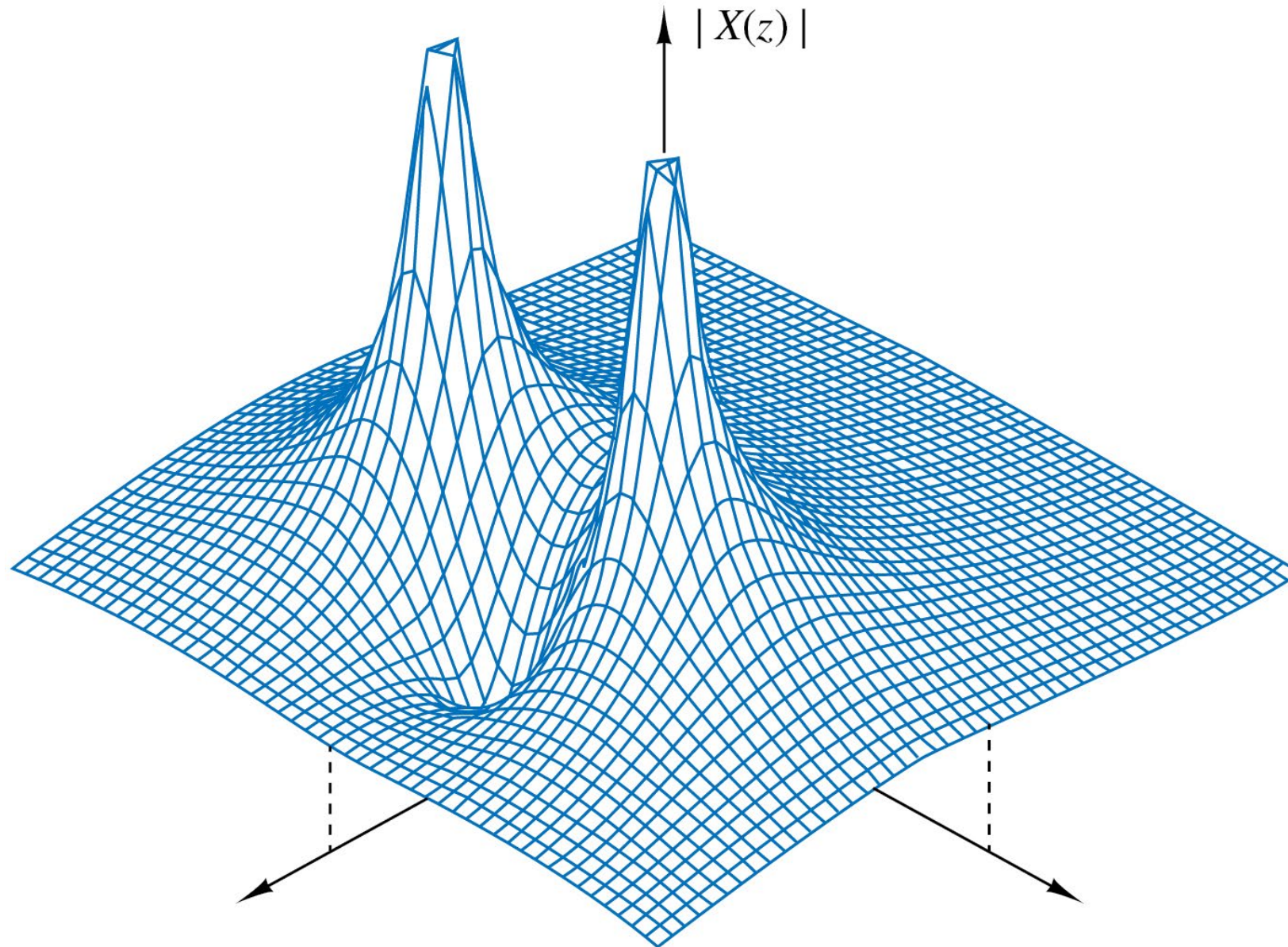


Figure 3.3.4 Graph of $|X(z)|$ for the z-transform in 3.3.3.



Causal Signals and Unit Circle

- The time behavior of a causal signal depends on the location of its poles relative to the unit circle $|z| = 1$.
- Example: A real signal with one pole:

$$x(n) = a^n u(n) \xleftrightarrow{z} X(z) = \frac{1}{1 - az^{-1}}, \quad \text{ROC: } |z| > |a|$$

$$z_1 = 0, p_1 = a$$

Figure 3.3.5 Time-domain behavior of a single-real-pole causal signal as a function of the location of the pole with respect to the unit circle.

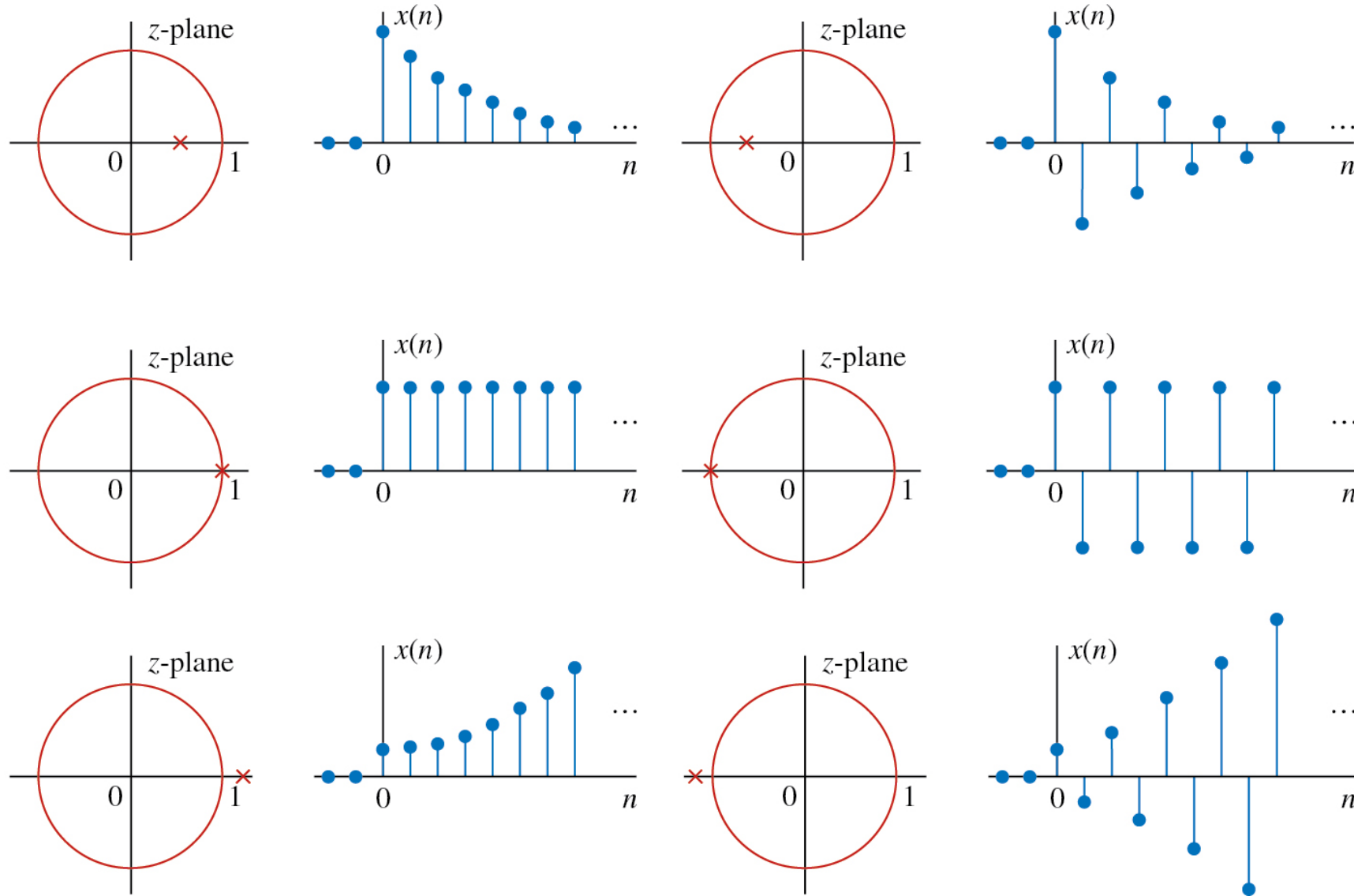


Figure 3.3.6 Time-domain behavior of causal signals corresponding to a double ($m = 2$) real pole, as a function of the pole location.

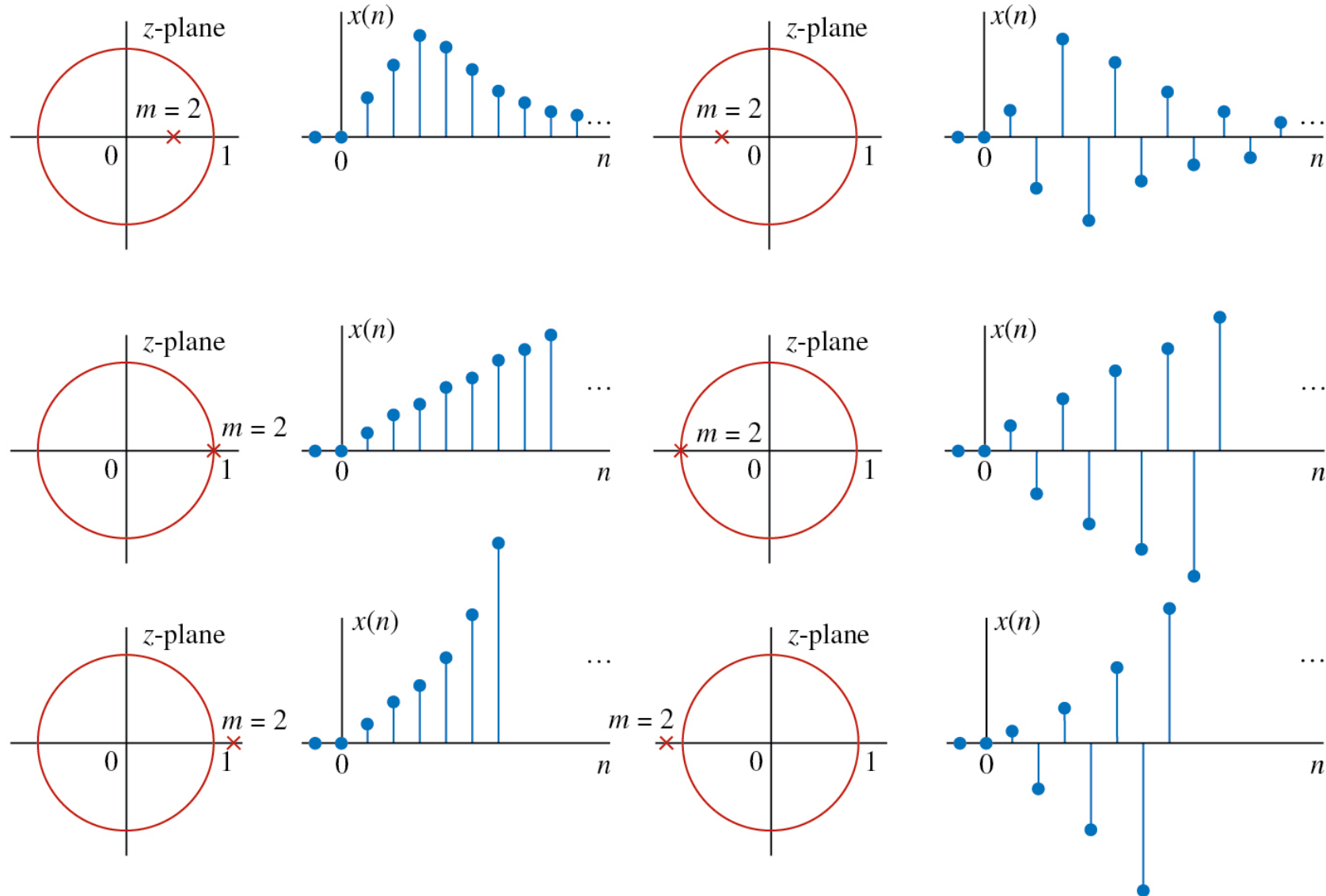


Figure 3.3.7 A pair of complex-conjugate poles corresponds to causal signals with oscillatory behavior.

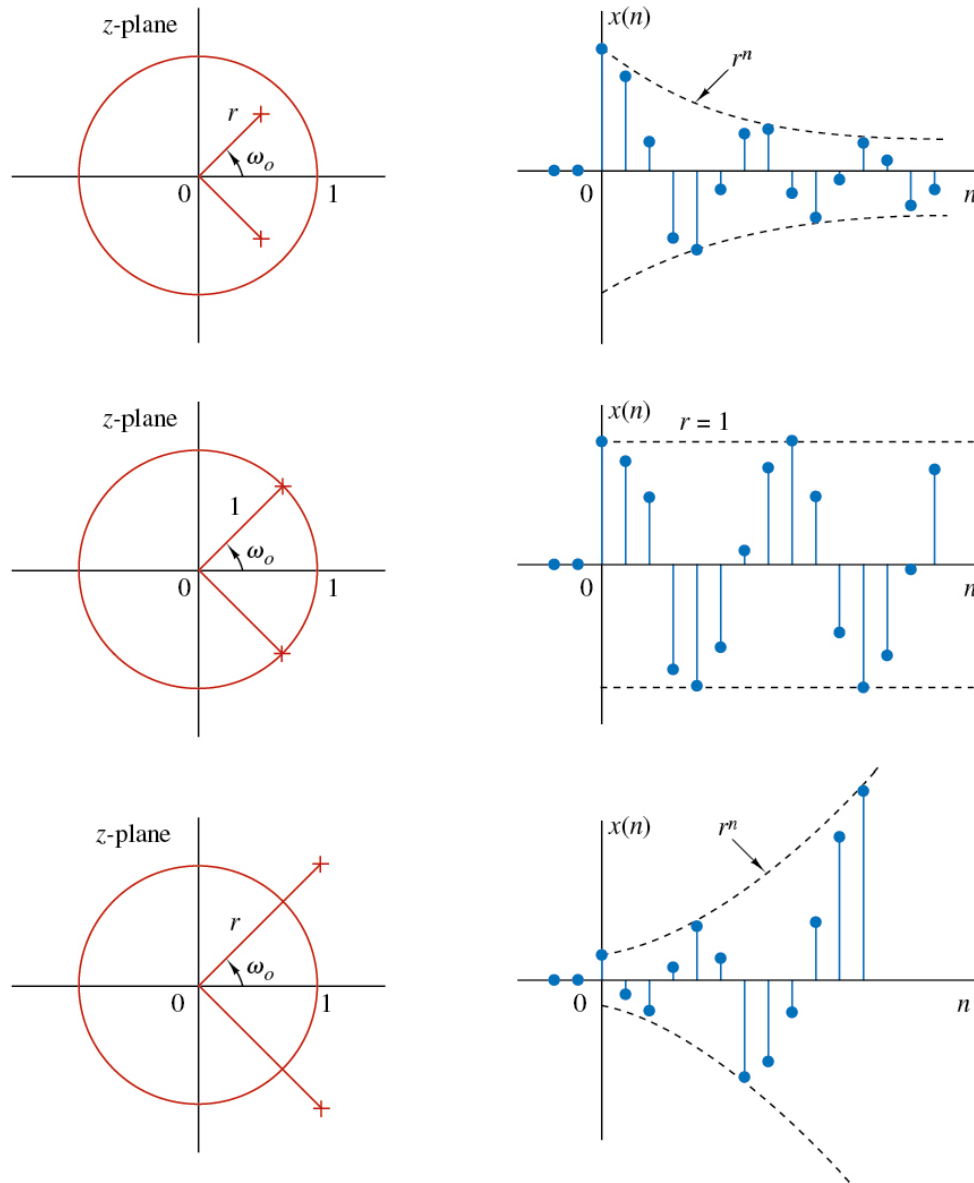
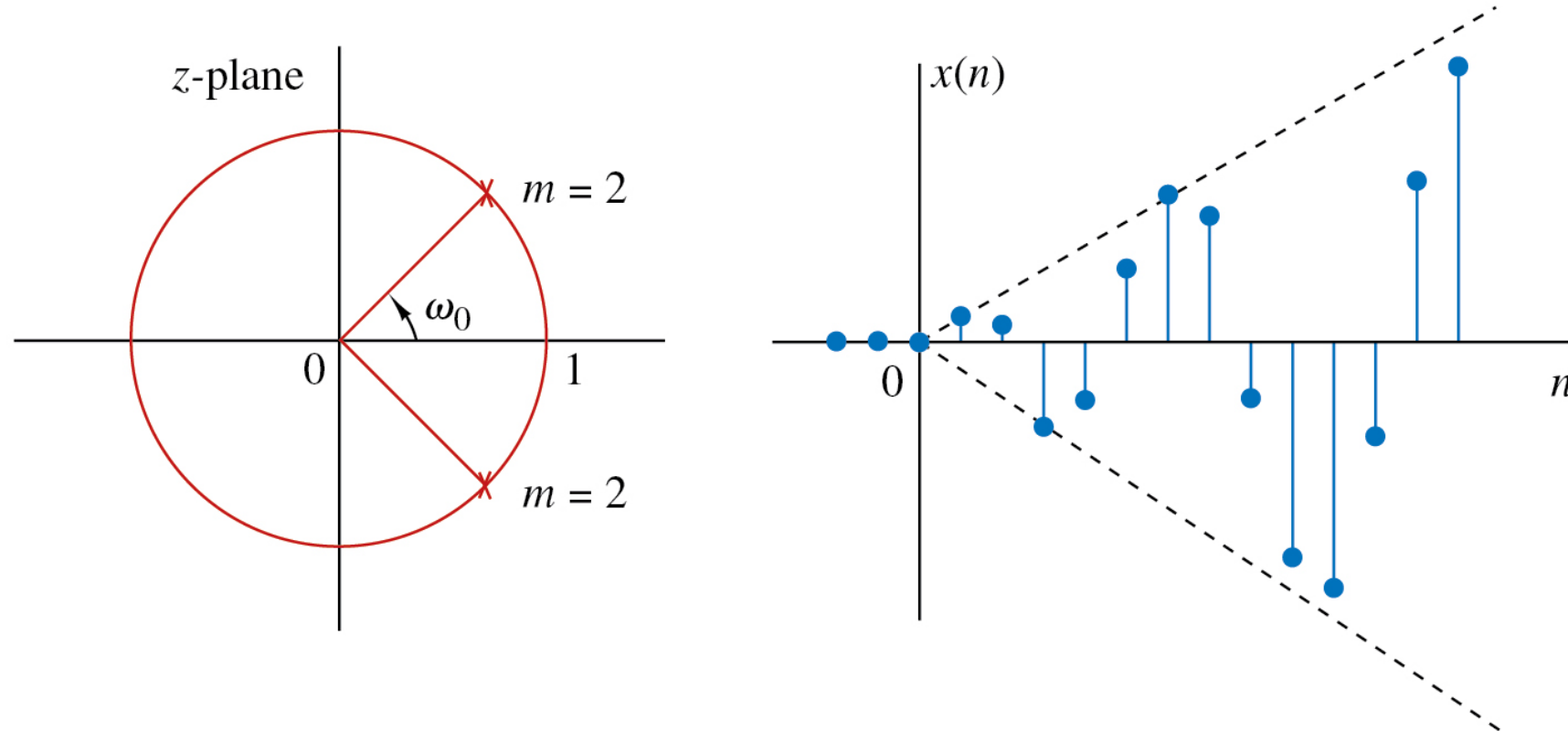


Figure 3.3.8 Causal signal corresponding to a double pair of complex-conjugate poles on the unit circle.



Pole Location and Time-Domain Behavior of Causal Signals

- The time behavior of a signal can be predicted based on the location of its poles relative to the unit circle.
- Causal real signals with simple real poles or simple complex-conjugate pairs of poles
 - If inside or on the unit circle, signals are always bounded in amplitude.
 - If a pole (or a complex-conjugate pair of poles) is near the origin, the signal decays faster.
- Zeros also affect the behavior of a signal
 - For example, the presence and location of zeros affects the phase of sinusoidal signals.

Pole Location and Behavior of Causal LTI Systems

- Since the impulse response of a causal LTI system is a causal signal, the pole location of the system determines whether the system can become unbounded/ unstable.
- **System function:** $H(z)$, the z-transform of the impulse response $h(n)$.

$$Y(z) = H(z)X(z)$$

- An LTI system described by a (constant-coefficient) difference equation has a **rational system function**.

Rational System Functions of LTI Systems

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$$Y(z) = - \sum_{k=1}^N a_k Y(z) z^{-k} + \sum_{k=0}^M b_k X(z) z^{-k}$$

$$Y(z) \left(1 + \sum_{k=1}^N a_k z^{-k} \right) = X(z) \left(\sum_{k=0}^M b_k z^{-k} \right)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

Rational System Functions

- **All-zero system:** If only trivial poles (at $z=0$), but M nontrivial zeros.
 - Finite-duration impulse response (FIR), or moving-average system
- **All-pole system:** If only trivial zeros (at $z=0$), but N nontrivial poles.
 - Infinite-duration impulse response (IIR)
- **Pole-zero system:** Nontrivial poles and zeros (in addition to trivial ones at $z=0$ or $z=\infty$)
 - Infinite-duration impulse response (IIR)

The Inverse z-Transform

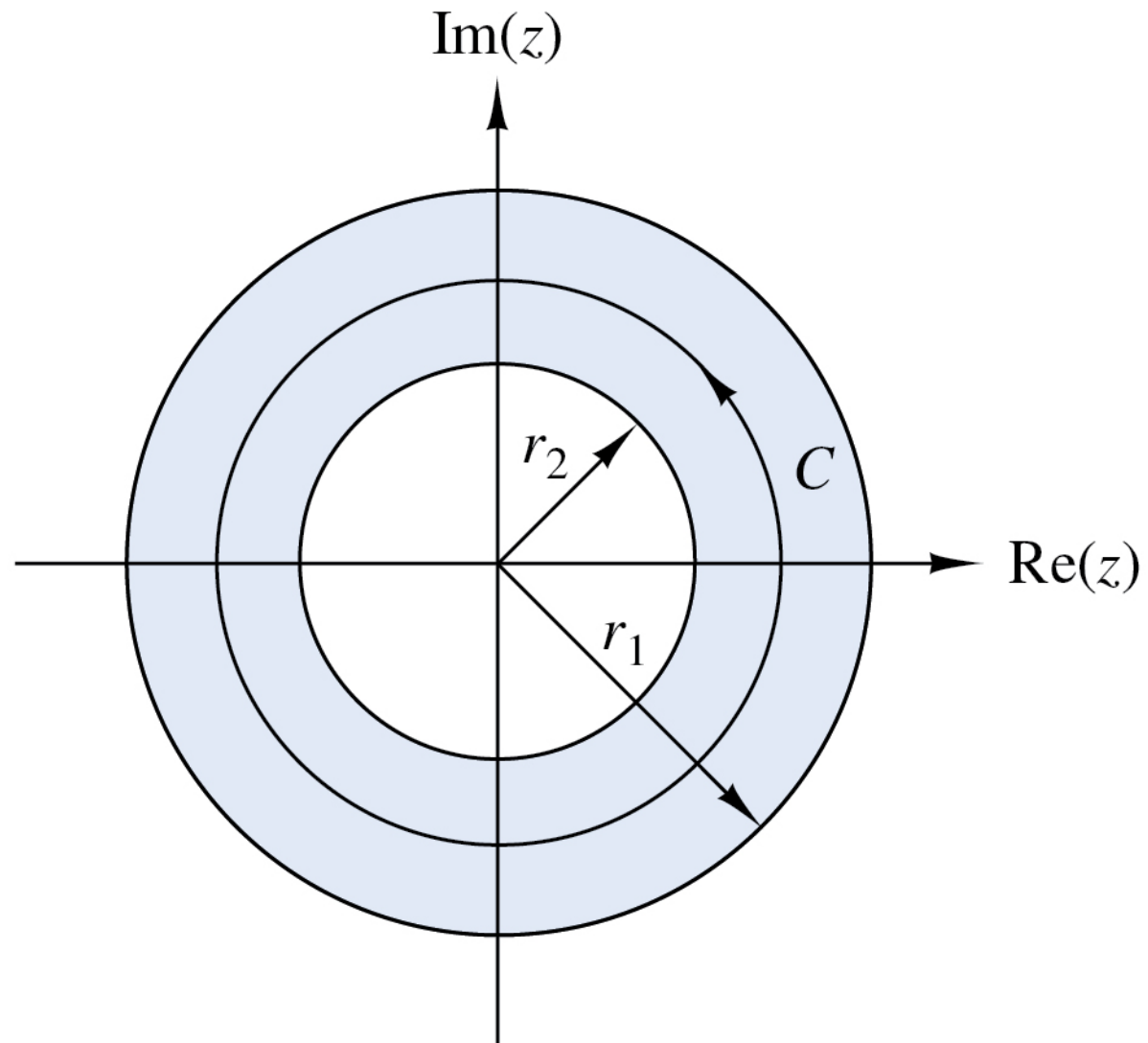
Inverse z-Transform

- Three methods:
 - Direct evaluation by contour integration

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

- Power series expansion
- Partial fraction expansion

Figure 3.1.5 Contour C for integral in (3.1.13)



Power Series Expansion Method

- Expand $Z(z)$ as a power series

$$X(z) = \sum_{n=-\infty}^{\infty} c_n z^{-n} \longleftrightarrow x(n) = c_n$$

- Causal signal if ROC is the exterior of a circle
- Anticausal signal if ROC is the interior of a circle
- Useful method for signals of short sequence, or to find the first few samples.

Linearity Property

- Recall, if

$$X(z) = \alpha_1 X_1(z) + \alpha_2 X_2(z) + \cdots + \alpha_K X_K(z)$$

And inverse transforms of $X_1(z)$, ..., $X_K(z)$ are $x_1(n)$, ..., $x_K(n)$, then

$$x(n) = \alpha_1 x_1(n) + \alpha_2 x_2(n) + \cdots + \alpha_K x_K(n)$$

Partial-Fraction Expansion Method

- Given the following (from the z-transform table):

$$Z^{-1}\left\{\frac{1}{1 - p_k z^{-1}}\right\} = \begin{cases} (p_k)^n u(n), & \text{if ROC: } |z| > |p_k| \\ & \text{(causal signals)} \\ -(p_k)^n u(-n - 1), & \text{if ROC } |z| < |p_k| \\ & \text{(anticausal signals)} \end{cases}$$

- The inverse z-transform of $X(z)$ with the following (partial fraction) form:

$$X(z) = A_1 \frac{1}{1 - p_1 z^{-1}} + A_2 \frac{1}{1 - p_2 z^{-1}} + \cdots + A_N \frac{1}{1 - p_N z^{-1}}, \quad \text{ROC } |z| > p_{\max}$$

- is a causal signal: $x(n) = (A_1 p_1^n + A_2 p_2^n + \cdots + A_N p_N^n) u(n)$

Expansion into Partial-Fraction Form

1. Write as a proper rational function ($N > M$)

$$X(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

2. Eliminate negative powers.

$$X(z) = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{z^N + a_1 z^{N-1} + \dots + a_N}$$

3. Find roots of denominator (poles).

$$\frac{X(z)}{z} = \frac{A_1}{z - p_1} + \frac{A_2}{z - p_2} + \dots + \frac{A_N}{z - p_N}$$

4. Find coefficients (A_1 to A_n) to rewrite as partial-fraction form.

If distinct poles:

$$A_k = \left. \frac{(z - p_k) X(z)}{z} \right|_{z=p_k}, \quad k = 1, 2, \dots, N$$

Proper Rational Function

$$X(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

- If $M < N$ (and $a_N \neq 0$), the above form is called a proper rational function.
- Otherwise, improper.
- Any improper rational function can always be written as the sum of a polynomial and a proper rational function.

$$X(z) = \frac{B(z)}{A(z)} = c_0 + c_1 z^{-1} + \dots + c_{M-N} z^{-(M-N)} + \frac{B_1(z)}{A(z)}$$

Other useful inverse z-transforms

$$Z^{-1} \left(\frac{A_k}{1 - p_k z^{-1}} + \frac{A_k^*}{1 - p_k^* z^{-1}} \right) = 2 |A_k| r_k^n \cos (\beta_k n + \alpha_k) u(n)$$

if the ROC is $|z| > |p_k| = r_k$.

$$Z^{-1} \left\{ \frac{p z^{-1}}{(1 - p z^{-1})^2} \right\} = n p^n u(n)$$

provided that the ROC is $|z| > |p|$.

Analysis of LTI Systems in z-Domain

Rational System Functions

- For a relaxed system, if $H(z) = B(z) / A(z)$ and $X(z) = N(z) / Q(z)$, then

$$Y(z) = H(z)X(z) = \frac{B(z)N(z)}{A(z)Q(z)}$$

- If p_i and q_i are distinct poles of $H(z)$ and $X(z)$ respectively,

$$Y(z) = \sum_{k=1}^N \frac{A_k}{1 - p_k z^{-1}} + \sum_{k=1}^L \frac{Q_k}{1 - q_k z^{-1}}$$

$$y(n) = \underbrace{\sum_{k=1}^N A_k (p_k)^n u(n)}_{\text{natural response}} + \underbrace{\sum_{k=1}^L Q_k (q_k)^n u(n)}_{\text{forced response}}$$

= natural response + forced response

Transient vs. Steady-State Responses

- $$y(n) = \underbrace{\sum_{k=1}^N A_k(p_k)^n u(n)}_{\text{natural response}} + \underbrace{\sum_{k=1}^L Q_k(q_k)^n u(n)}_{\text{forced response}}$$
- natural response + forced response
- If $|p_k| < 1$ for all k , then the *natural response* decays to zero at infinity, and is therefore, referred to as the **transient response**.
- If the causal input signal is sinusoidal, then the *forced response* is also sinusoidal and persists for all $n > 0$ and is thus referred to as the **steady-state response**.

Causality and Stability

- An LTI system is **causal** if and only if the ROC of the system function is the exterior of a circle of radius $r < \infty$, including the point $z=\infty$.
- An LTI system is BIBO **stable** if and only if the ROC of the system function includes the unit circle.
- A causal and stable LTI system must have a system function that converges for $|z| > r < 1$. A causal LTI system is BIBO stable if and only if all the poles of $H(z)$ are inside the unit circle.

Stability of Second-Order System

- A Causal two-pole system is described by the second-order difference equation:

$$y(n) = -a_1y(n-1) - a_2y(n-2) + b_0x(n)$$

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \frac{b_0}{1 + a_1z^{-1} + a_2z^{-1}} \\ &= \frac{b_0z^2}{z^2 + a_1z + a_2} \end{aligned}$$

$$p_1, p_2 = -\frac{a_1}{2} \pm \sqrt{\frac{a_1^2 - 4a_2}{4}}$$

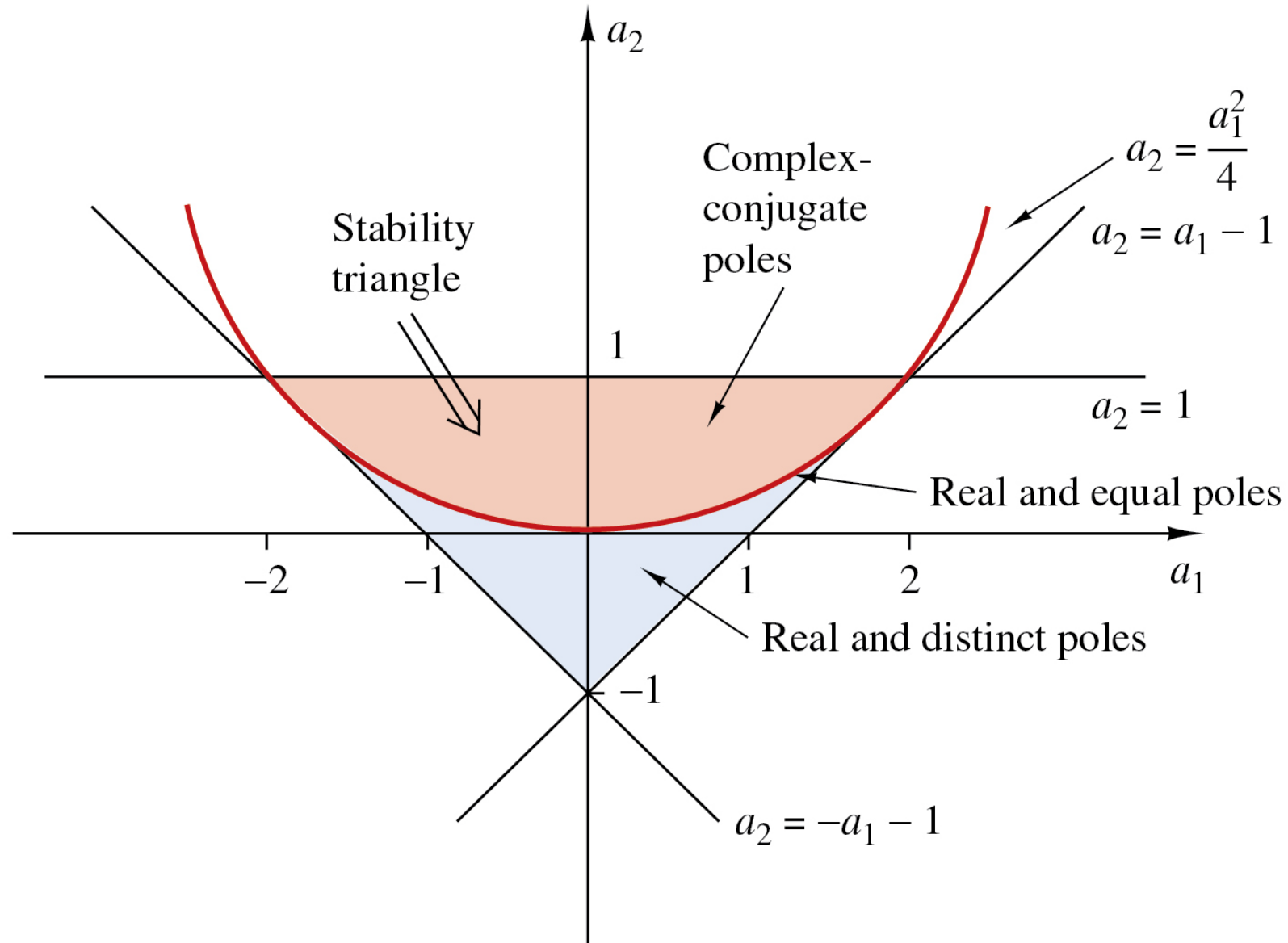
- BIBO stable if poles inside unit circle, therefore

$$|a_2| = |p_1p_2| = |p_1||p_2| < 1$$

$$|a_1| < 1 + a_2$$

- Stability Triangle

Figure 3.5.1 Region of stability (stability triangle) in the (a_1, a_2) coefficient plane for a second-order system.



Example: Real and Distinct poles

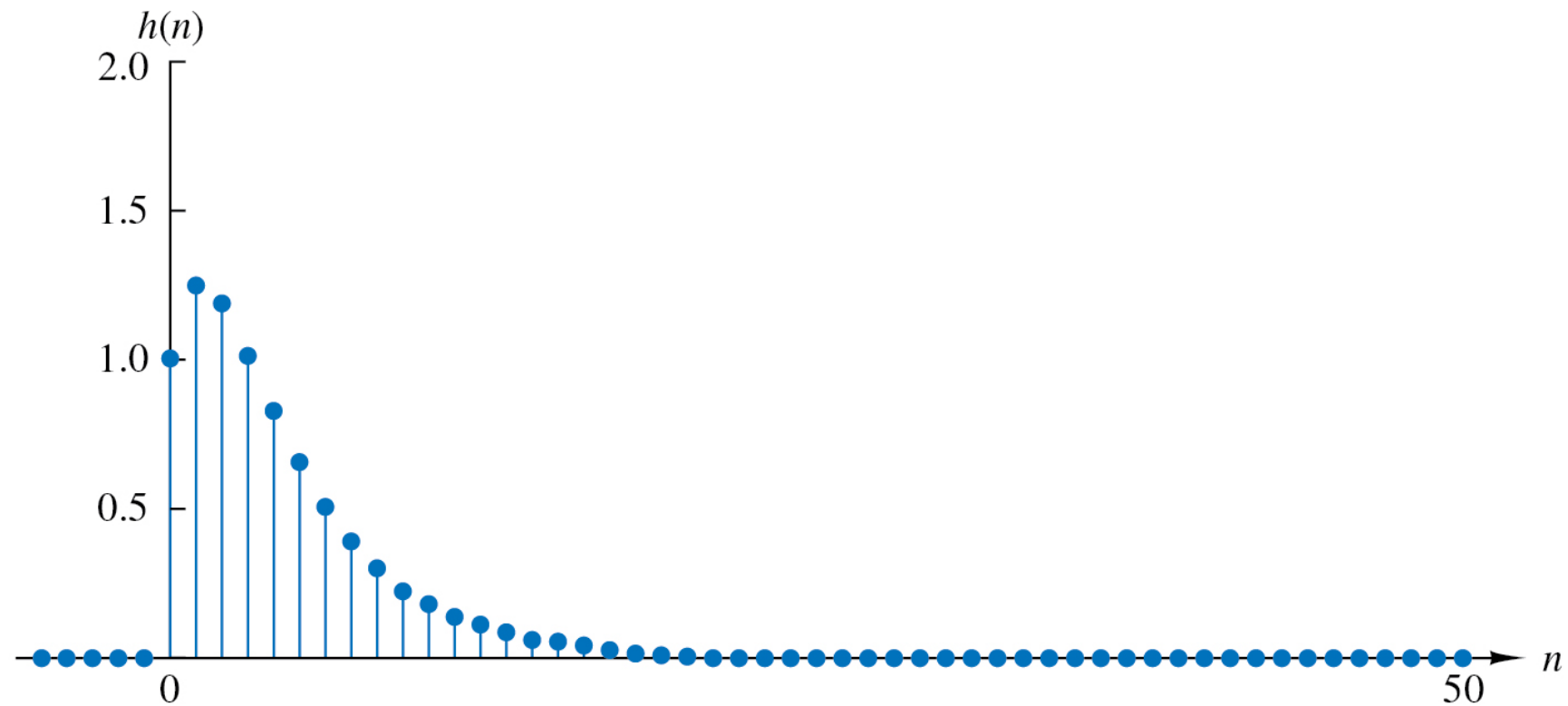
- When real and distinct poles ($a_1^2 > 4 a_2$), the unit sample response is the difference of two decaying exponential sequences.

$$H(z) = \frac{A_1}{1 - p_1 z^{-1}} + \frac{A_2}{1 - p_2 z^{-1}}$$

$$A_1 = \frac{b_0 p_1}{p_1 - p_2}, \quad A_2 = \frac{-b_0 p_2}{p_1 - p_2}$$

$$h(n) = \frac{b_0}{p_1 - p_2} (p_1^{n+1} - p_2^{n+1}) u(n)$$

Figure 3.5.2 Plot of $h(n)$ given by (3.5.16) with $p_1 = 0.5$, $p_2 = 0.75$;
 $h(n) = [1/(p_1 - p_2)](p_1^{n+1} - p_1^{n+2})u(n)$.



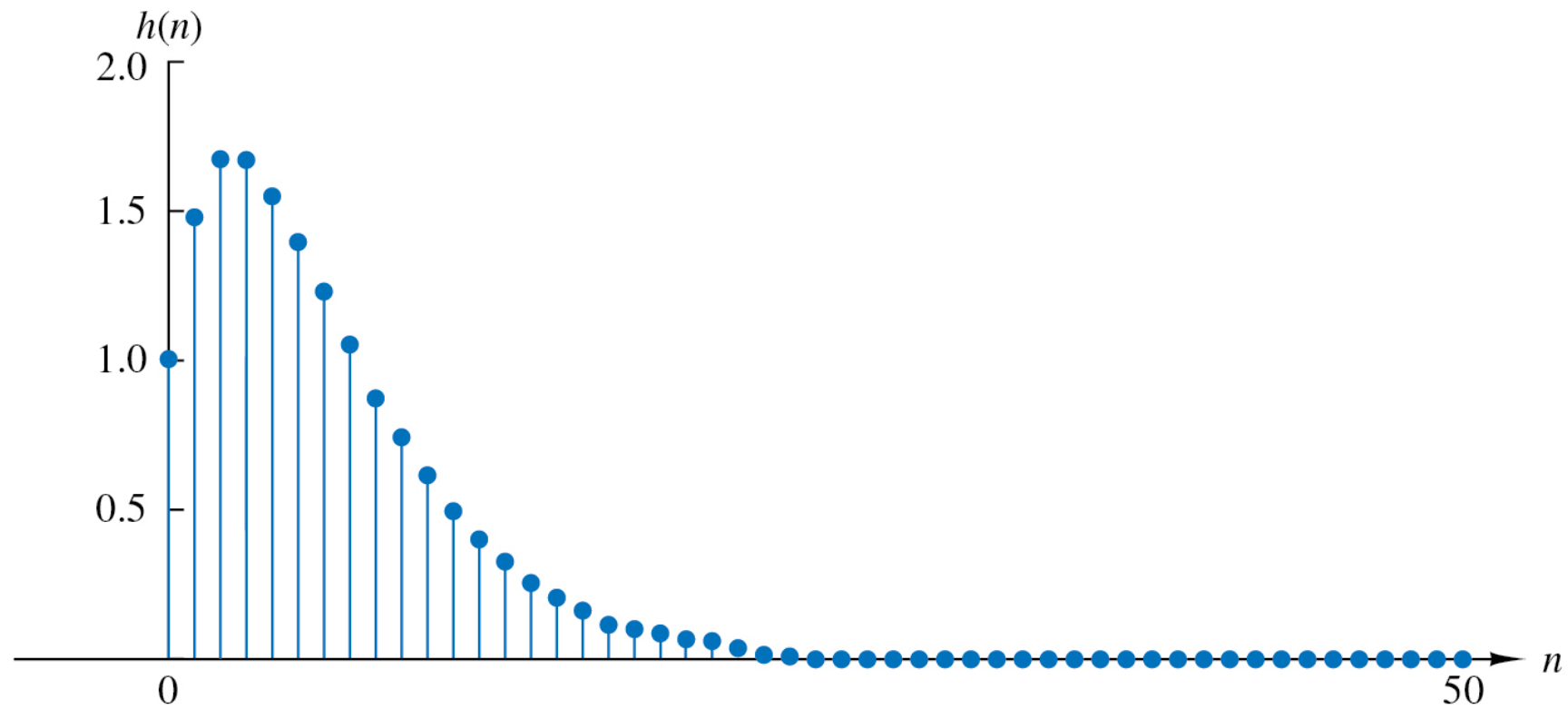
Example: Real and equal poles

- When real and distinct poles ($a_1^2 = 4 a_2$), the unit sample response is the product of a ramp sequence and a real decaying exponential sequence.

$$H(z) = \frac{b_0}{(1 - pz^{-1})^2}$$

$$h(n) = b_0(n + 1)p^n u(n)$$

Figure 3.5.3 Plot of $h(n)$ given by (3.5.18) with $p = \frac{3}{4}$; $h(n) = (n + 1) p^n u(n)$.



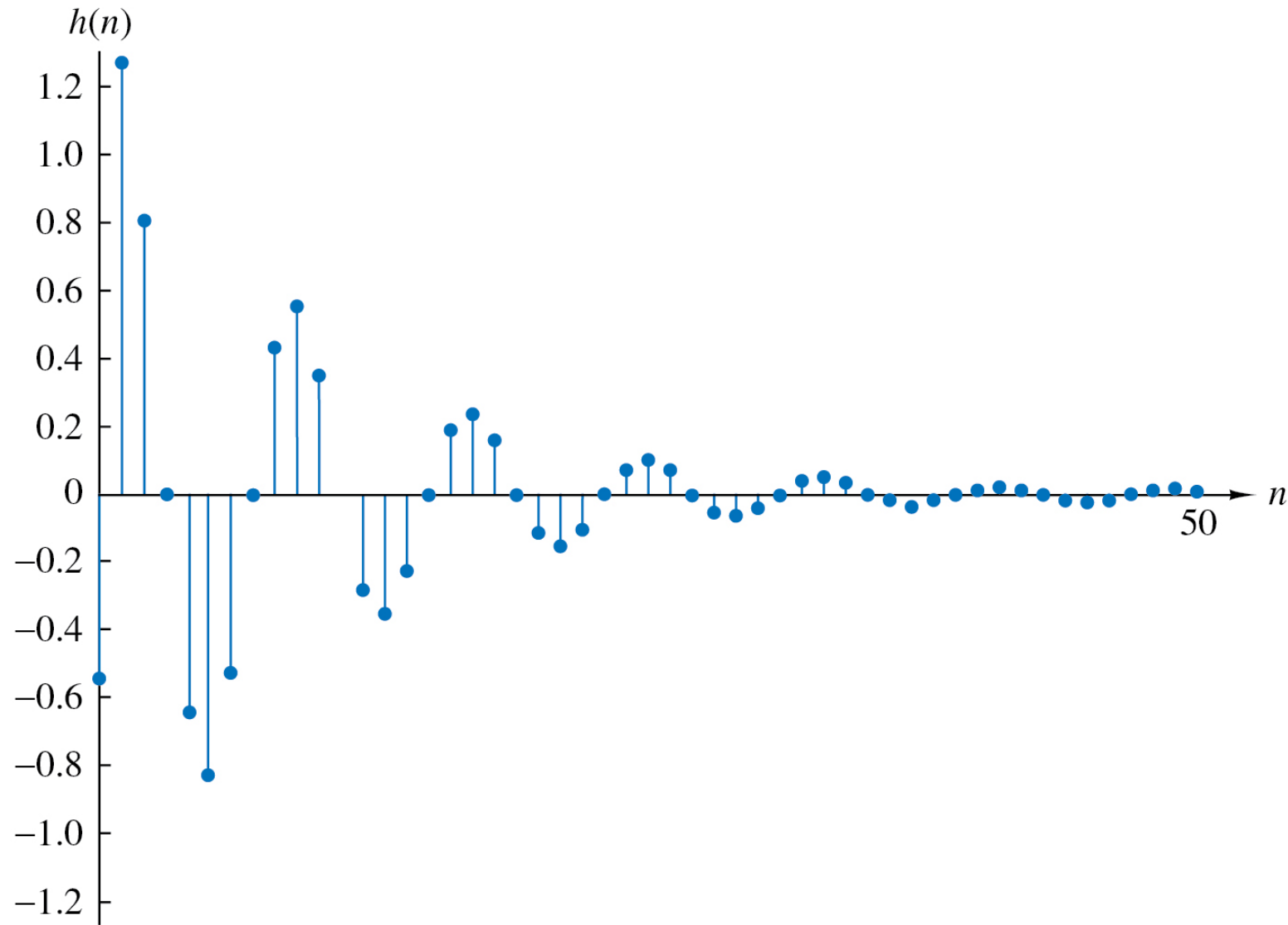
Example: Complex-conjugate poles

- When real and distinct poles ($a_1^2 < 4 a_2$), the unit sample response has an oscillatory behavior with an exponentially decaying envelope when $r < 1$.
- ω_0 (angle of poles): determines the frequency of oscillation.
- r (distance of poles from the origin): determines the rate of decay.

$$\begin{aligned} H(z) &= \frac{A}{1 - pz^{-1}} + \frac{A^*}{1 - p^*z^{-1}} \\ &= \frac{A}{1 - re^{j\omega_0}z^{-1}} + \frac{A^*}{1 - re^{-j\omega_0}z^{-1}} \end{aligned}$$

$$\begin{aligned} h(n) &= \frac{b_0 r^n}{\sin \omega_0} \frac{e^{j(n+1)\omega_0} - e^{-j(n+1)\omega_0}}{2j} u(n) \\ &= \frac{b_0 r^n}{\sin \omega_0} \sin (n+1)\omega_0 u(n) \end{aligned}$$

Figure 3.5.3 Plot of $h(n)$ given by (3.5.22) with $b_0 = 1$, $\omega_0 = \pi/4$, $r = 0.9$; $h(n) = [b_0 r^n / (\sin \omega_0)] \sin[(n + 1)\omega_0] u(n)$.



Summary

References

- [DSP] Proakis, J.G. and Manolakis, D.G., Digital Signal Processing: Principles, Algorithms, and Applications, Pearson, 5th ed., 2022.

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